
MiSTer Monitor

Setup Guide

v2.10.0

RETROACHIEVEMENTS

PROGRESS 23/40

57%



[H]	Perfect Run	25p
[*]	First Blood	10p
[]	Speed Demon	50p
[]	No Continues Used	25p

ACHIEVEMENT!

+25 pts Perfect Run



What is MiSTer Monitor

MiSTer Monitor turns an ESP32-based touchscreen into a companion display for your MiSTer: the moment you load a game, the screen shows its box art, a “Now Playing” panel with related info (year, dev, publisher, genre, players, rating and synopsis) of the active game, your **RetroAchievements** progress, and live system stats. Everything is fetched on demand.

Two pieces work together: a tiny **server** on the MiSTer that watches what you play, and the **firmware** on the display that renders it. You will install both.

You will need

- A **MiSTer** connected to your home network.
- A supported display: **M5Stack Tab5** (5"), **Guition JC8012P4A1C** (10.1"), or a **Cheap Yellow Display** (2.8" ILI9341/ST7789, or 3.5" capacitive/resistive).
- A **microSD card** for the display.
- A **USB data cable** and a desktop computer with **Chrome or Edge** (only for the one-time flash).
- A free **ScreenScraper** account — created in step 3.
- Optional: a free **RetroAchievements** account — step 5.

1

Flash the display

- Connect the display to your computer with the USB cable.
- Open the flasher page in **Chrome or Edge** (Firefox and Safari don't support Web Serial):

```
https://chipster6502.github.io/MiSTer\_monitor/flasher/
```

- Click **Connect** next to your board, pick the serial port in the browser dialog, and wait for the confirmation.
- Not sure which CYD variant you have? The page itself explains how to tell them apart.

2

Prepare the display's microSD card

The repository ships a ready-made card layout under **SD_card_content/**. Pick the subfolder matching your board, copy its contents to the **root** of the microSD card, then open **config.ini** in any text editor and fill in your network — **without quotes** around the values:

```
config.ini – on the display's microSD card
[wifi]
ssid=YOUR_WIFI_NAME
password=YOUR_WIFI_PASSWORD
```

That is all the display needs to find the MiSTer: it discovers it automatically on the network at boot.

This first edit has to happen on a computer, because the display needs the WiFi details before it can be reached over the network. From then on you can edit **config.ini** from a browser instead — see step 6.

NOTE Leave **[mister] ip=** blank. Only set it if your router blocks UDP broadcast, which is rare in home networks.

3 Create a free ScreenScraper account

- Register at screenscraper.fr — free, instant, no approval needed.
- Add your username and password to the same **config.ini**:

```
config.ini – same file, next section
[screenscraper]
ss_user=YOUR_SS_USERNAME
ss_pass=YOUR_SS_PASSWORD
```

The app brings its own developer credentials, so a plain member account is enough. Artwork is downloaded the first time each game appears and cached on the card, so it costs quota once and loads instantly forever after.

4 Install the server on the MiSTer

One toggle in Update All or MiSTer Companion — it updates itself from then on

Pick whichever updater you already use. Both install the same files and keep them current from then on.

- **Update All** — run it from the Scripts menu and press **UP** during the countdown to open the settings. Go to **Tools & Scripts** and enable **MiSTer Monitor**, then save and let the update finish. (Requires Update All 2.10 or newer.)
- **MiSTer Companion** — open **Install Center**, find **MiSTer Monitor** and install it.
- Back in the Scripts menu, run **MiSTer_Monitor** once. It enables auto-start, fixes **MiSTer.ini** for game detection, and starts the server right away. It is safe to run again at any time.
- Insert the card in your display and power it.
- Load any game. Within a few seconds the display shows its artwork.

TIP While you are in the Update All settings, look under **Extra Content** → **Game Artwork DBs**. The [artwork packs](#) install box art for thousands of games straight onto the MiSTer, so covered games appear instantly without a ScreenScraper lookup. They are optional — everything works without them, and ScreenScraper still fills in whatever a pack does not cover.

NOTE Running the **Downloader** on its own, without Update All? A drop-in database file does the same job — see [installation.md](#). Use one route or the other, not both.

IMPORTANT **Console Mode** (Taki Udon's frontend) replaces **MiSTer.ini** with its own copy, which turns game detection back off. If you install or repair it, run **MiSTer_Monitor** once more afterwards and everything returns to normal.

To pause the monitor, run **MiSTer_Monitor_uninstall** from the Scripts menu; run **MiSTer_Monitor** again to re-enable it.

5

RetroAchievements (optional)

Your progress and live unlock popups on the display

This part lives **on the MiSTer**, not on the display's card. The first time the server runs it drops a ready-to-edit file in place — you only fill in two lines:

```
on the MiSTer – created automatically on first start
/media/fat/Scripts/.config/mister_monitor/ra_credentials.ini

[retroachievements]

username=YourRAUsername

api_key=YourWebAPIKey
```

Your **Web API key** is at retroachievements.org → *Settings* → *Applications*. Load a game with a published set and the RetroAchievements page comes alive: progress bar, points, and the full trophy list — tap any row for its description.

Your setup	What you get
Credentials only stock MiSTer	The full page in view-only mode: the game's set, your cloud progress and every trophy — but play is not recorded.
odelot/Main_MiSTer fork RA_cores	Achievements are earned while you play ; a READY banner confirms it on each matched game, and unlocks pop within seconds.
Fork + debug=1 retroachievements.cfg	Popups become instant (with title and description), and CD systems — PlayStation, Saturn, Mega CD — resolve too.

IMPORTANT Each tier is optional and falls back gracefully to the one below. Without the credentials file, everything else in this guide works exactly the same.

Earning achievements on the MiSTer

Recording achievements needs odelot's RetroAchievements build of the MiSTer main binary, plus his modified core for each system. Two maintained routes install it for you, both keeping your stock cores intact:

- **Update All** — the RetroAchievements Cores database, maintained by odelot himself, is now built in. Run **Update All**, enable it in the settings screen, and it walks you through the `retroachievements.cfg` setup and the `MiSTer.ini` entries it needs.

- **MiSTer Companion** — install the **RetroAchievements Cores** entry from its **Install Center**. (mistercompanion.org)

Either route leaves a `retroachievements.cfg` on the MiSTer holding your RA **username and password**. That file belongs to odelot's build and is separate from the `ra_credentials.ini` above, which is this project's and takes an API key instead — you will end up with both, and that is expected.

```
two different files, both needed for the full experience
/media/fat/retroachievements.cfg      odelot's build (user + password)
/media/fat/Scripts/.config/mister_monitor/ra_credentials.ini
                                     this project   (user + API key)
```

IMPORTANT Once the RA cores are installed an **RA_Cores** entry appears in the MiSTer OSD. Achievements are only active for cores launched from there — opening the same system from its usual menu entry runs the stock core and records nothing.

TIP Add `debug=1` to `retroachievements.cfg` for sub-second unlock popups. Achievements only work on the systems odelot has published cores for — [his repository](#) lists the current set.

6

The web interface

Edit settings and manage the display's card from any browser

Once the display is on your network it runs a small web interface of its own. Point a browser at **the display's own IP address** on port 8080 — not the MiSTer's. The address is printed on the serial console at boot and appears in your router's device list.

```
edit config.ini
http://<display-ip>:8080/config
```

The page opens the card's **config.ini** in a text editor, exactly as it is written, comments and all. Saving writes it back to the microSD and keeps the previous version as **config.ini.bak** on the same card. Settings are read once at startup, so the page offers a **Reboot** button to apply them — the display restarts in a couple of seconds.

```
browse the microSD card
http://<display-ip>:8080/files
```

The same interface lists the card's contents: open folders, download a file to your computer, upload new ones, create folders, and delete what you no longer need. The most common use is artwork — delete the image for one game and it will be fetched again next time you load it, or upload your own to replace it.

NOTE Artwork you upload must be **baseline** JPEG. Progressive JPEGs are rejected with a message, because the display's decoder cannot read them. Most image editors offer the choice when exporting. Deleting covers files and empty folders; to remove a folder with contents, empty it first.

The interface has no password by default, which is fine on a home network. To lock it, set a password in the **[ui]** section of config.ini — the browser will then ask for it, with the username **admin**. Setting **web_config=false** disables the whole web interface.

```
config.ini — optional
[ui]
web_config=true
web_password=
```

The 2.8" boards, the Tab5 and the Guition also serve a screenshot of what is on screen at **http://<display-ip>:8080/**. The 3.5" boards do not: their panel cannot be read back.



You're done — a quick check

Working as intended looks like this

- Boot: the display connects to WiFi and finds the MiSTer by itself.
- Load a game: box art appears within seconds; the Game Info panel fills in year, developer and synopsis.
- RetroAchievements page: progress and trophies for the running game (view-only unless you installed the fork and open a RA enabled core).
- System pages: live CPU, memory, storage and network stats.
- Web interface: **http://<display-ip>:8080/config** opens the settings editor from any browser on your network.

```
links
Project      github.com/chipster6502/MiSTer_monitor
Docs         installation.md  configuration.md
Flasher      chipster6502.github.io/MiSTer_monitor/flasher/
RetroAchiev. retroachievements.org
RA fork      github.com/odelot/Main_MiSTer
Update All   github.com/theypsilon/Update_All_MiSTer
Companion    mistercompanion.org
Display UI    http://<display-ip>:8080/config
Server page  http://<mister-ip>:8081/
```

MiSTer Monitor is free software — firmware under GPL-3.0, server under AGPL-3.0. Found a rough edge in this guide? Open an issue; it helps the next person.